Linkai Peng

Ph.D. Student, Language & Cognition · University of Connecticut

Audio language models · Human speech perception

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Education

University of Connecticut, Ph.D., Language & Cognition · Storrs, CT — Sep 2024 – 2029 (expected)
- Computational models of human speech recognition

Beijing Language and Culture University, M.S., Software Engineering (SAIT Lab) · Beijing — Sep 2019 – Jun 2022
- Speech recognition systems for second-language speakers

South China Normal University, B.S., Physics · Guangzhou — Sep 2015 – Jun 2019

Industry Experience

NetEase (Youdao), Senior Machine Learning Engineer, Speech · Beijing — Jul 2022 – Jul 2024
- Real-time and on-device speech systems for interactive language learning (see Hi Echo).

ByteDance AI Lab, Research Intern, Speech · Beijing — Nov 2020 – Feb 2021
- ASR modeling for pronunciation scoring.

Selected Publications

Full list: Google Scholar

Full-Duplex Speech Models Take the Floor When Asked, Not When Needed
L. Peng, B. Nuchged, K. Fu, Y. Yao · *arXiv 2609.19596, under review*
Benchmark built on human turn-taking rules: models rarely take the floor when needed (e.g., hazards, false claims).

Listen, Diagnose, and Interrupt: Coupling Mispronunciation Diagnostics into a Full-Duplex Speech LLM
L. Peng, B. Nuchged, K. Fu, Y. Yao, J. Zhang · *Under review, 2026*
A full-duplex tutor that corrects mispronunciations mid-utterance (median **516 ms**) without losing detection performance.

Beyond the Transcript: Measuring and Improving the Use of Acoustic Information in Audio Language Models
L. Peng, B. Nuchged, Y. Yao · *Under review, 2026*
Audio LLMs do well when the words suffice but gain little from how they were said (e.g., emotion, stress). We measure this gap and lift Qwen2.5-Omni and encoder-free Gemma-4 12B by **12.8–44.4 pt**, with Qwen close to Gemini 3.1 Pro.

Do Machines Listen Like Humans? A Temporal Benchmark for Phonological Competition in End-to-End ASR
L. Peng, C. Brodbeck, S. Luthra, K. Brown, J. Rueckl, M. Escabí, D. Gow, J. S. Magnuson · *Interspeech 2026* · code
Causality makes a model's cognitive dynamics human-like. Since humans accumulate acoustic evidence over time, overconfidence and hallucination in speech models may stem from non-causal encoders (audio Jev may help).

A Study on Fine-Tuning wav2vec2.0 Model for the Task of Mispronunciation Detection and Diagnosis
L. Peng, K. Fu, B. Lin, D. Ke, J. Zhang · *Interspeech 2021*
Fine-tuning wav2vec 2.0 reaches **60.44%** F1 on L2-ARCTIC, and 55.52% with 3× less labelled data.

Skills

Speech Engineering: ASR (wav2vec 2.0, WavLM, Whisper) · Speech LLMs (Moshi, PersonaPlex, VoiceChat, Qwen2.5-Omni, Gemma-4) · Python, PyTorch, C++
Cognitive Science: visual-world eye tracking, experiment design, human benchmarks for speech models

Honors

Shankweiler Fellow, Donald Shankweiler Language Sciences Award, CNC-CT program, University of Connecticut

Outstanding Graduation Dissertation Award, Beijing Language and Culture University — 2022

Service

Reviewer, ICASSP 2025, 2026, 2027

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